Uniqueness of order isomorphisms between well-orderings

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the induction framing, the full one-direction transport x lessdot f(a) implies x lessdot g(a), the symmetric remark, and the external named citation are verbatim in the span.

Editorial note. The label ordisounique names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Well-ordering induction. Same unit, \ollabel{propwoinduction} (source line 1265). The proof's opening (line 749) and closing (line 761) induction: to show f = g it suffices to propagate equality along initial segments of A.
- Extensionality for strict linear orders. Cross-unit citation \olref[sfr][rel][ord]{prop:extensionality-strictlinearorders} invoked at line 760; the target unit (Open Logic Project, Sets/Functions/Relations: Relations) is not staged in this corpus. Two elements with the same strict-lower-set are equal: converts the pointwise equivalence x lessdot f(a) iff x lessdot g(a) into f(a) = g(a).

Source reference. Original proposition prop-extensionality-strictlinearorders: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 760–760. Named citation invoked at source line(s) 760: the author's marker \olref[sfr][rel][ord]{prop:extensionality-strictlinearorders} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition propwoinduction: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 1265–1265. Prerequisite pointer invoked at proof line(s) 749; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If $\langle A, <\rangle$ and $\langle B, \lessdot\rangle$ are isomorphic well-orderings, then the isomorphism between them is unique.*

Proof. Let f and g be isomorphisms $A \to B$. We will prove the result by induction, i.e. using propwoinduction. Fix $a \in A$, and suppose (for induction) that $(\forall b < a) f(b) = g(b)$. Fix $x \in B$.

If $x \lessdot f(a)$, then $f^{-1}(x) < a$, so $g(f^{-1}(x)) \lessdot g(a)$, invoking the fact that f and g are isomorphisms. But since $f^{-1}(x) < a$, by our supposition $x = f(f^{-1}(x)) = g(f^{-1}(x))$. So $x \lessdot g(a)$. Similarly, if $x \lessdot g(a)$ then $x \lessdot f(a)$.

Generalising, $(\forall x \in B)(x \lessdot f(a) \leftrightarrow x \lessdot g(a))$. It follows that $f(a) = g(a)$ by prop-extensionality-strictlinearorders. So $(\forall a \in A) f(a) = g(a)$ by propwoinduction. □